EMT 1101 *Continuous Assessment Examination Test*

Friday, June 17, 2016 **Time:** $1\frac{1}{2}$ **hours**

1. Check that the paper has **4 questions**.
2. Attempt all questions
3. Show all your working clearly
4. Use proper notation
5. Simplify your answers as far as possible
6. **No Calculators**

1. Find all real solutions of each of the following equations

(a) $4|x-2| = 3x - 4$ (4)

Solution: Two cases to consider
Case 1: $x - 2$ negative
Then $|x-2| = -(x-2)$, so the equation becomes
$-4(x-2) = 3x - 4$
$-4x + 8 = 3x - 4$
$7x = 12$
$x = \frac{12}{7}$

Case 2: $x - 2$ positive
Then $|x-2| = (x-2)$, so the equation becomes
$4(x-2) = 3x - 4$
$4x - 8 = 3x - 4$
$x = 4$

(b) $1 + 14x^{-1} + 48x^{-2} = 0$ (4)

Solution: Multiplying through by x^2 we get
$x^2 + 14x + 48 = 0$
$(x+8)(x+6) = 0$
From which $x = -8$ or $x = -6$

(c) $y^{\frac{2}{3}} - 21y^{\frac{1}{3}} = -80$ (5)

Solution: Let $x = y^{\frac{1}{3}}$, then the equation becomes
$x^2 - 21x + 80 = 0$
$(x-16)(x-5) = 0$

From which $x = 16$ or $x = 5$
But $x = y^{\frac{1}{3}}$, so we get $y^{\frac{1}{3}} = 16$ or $y^{\frac{1}{3}} = 5$ from which $y = 16^3 = 4096$ or $y = 5^3 = 125$

(d) $\dfrac{\sqrt{x}-4}{\sqrt{x}-18} = 3$ (5)

Solution: $\frac{\sqrt{x}-4}{\sqrt{x}-18} = 3$
$\sqrt{x} - 4 = 3(\sqrt{x} - 18)$
$\sqrt{x} - 4 = 3\sqrt{x} - 54$
$2\sqrt{x} = 50$
$\sqrt{x} = 25$
$x = 625$

(e) $\dfrac{x}{5-x} = \dfrac{-2}{11-x}$ (5)

Solution: $\frac{x}{5-x} = \frac{-2}{11-x}$
$11x - x^2 = -10 + 2x$
$x^2 - 9x - 10 = 0$
$(x - 10)(x + 1) = 0$
From which $x = 10$ or $x = -1$

2. (a) Find a quadratic equation that has integer coefficients and the following roots; $\frac{1+\sqrt{5}}{3}$ and $\frac{1-\sqrt{5}}{3}$ (5)

Solution: $c = \left(\frac{1+\sqrt{5}}{3}\right)\left(\frac{1-\sqrt{5}}{3}\right) = \frac{1^2-5}{9} = -\frac{4}{9}$ and $b = -\left(\frac{1+\sqrt{5}}{3} + \frac{1-\sqrt{5}}{3}\right) = -\frac{2}{3}$.
Hence the equation is $x^2 - \frac{2}{3}x - \frac{4}{9} = 0$ which yields $9x^2 - 6x - 4 = 0$

(b) Determine the value(s) of the constant k for which the equation $x^2 + 2(k+1)x + k^2 = 0$ has equal roots. (4)

Solution: We find k for which $b^2 - 4ac = 0$
That is
$(2(k + 1))^2 - 4k^2 = 0$
$4k^2 + 8k + 4 - 4k^2 = 0$
$8k + 4 = 0$
$8k = -4$
$k = -\frac{1}{2}$

3. Solve the following inequalities leaving your answer in interval notation

(a) $\dfrac{3x}{5} - \dfrac{(x-1)}{3} < 1$ (4)

Solution: $\frac{3x}{5} - \frac{(x-1)}{3} < 1$
$9x - 5(x-1) < 15$
$9x - 5x + 5 < 15$
$4x < 10$
$x < \frac{5}{2}$
Hence the solution set is $(-\infty, \frac{5}{2})$

(b) $\left|\frac{x+1}{2} - \frac{(x-1)}{3}\right| < 1$ (5)

Solution:
$\left|\frac{x+1}{2} - \frac{(x-1)}{3}\right| < 1$ implies $-1 < \frac{x+1}{2} - \frac{(x-1)}{3} < 1$
$-6 < 3(x+1) - 2(x-1) < 6$
$-6 < 3x + 3 - 2x + 2 < 6$
$-6 < x + 5 < 6$
$-11 < x < 1$
Hence the solution set $(-11, 1)$

(c) $\frac{x^2-1}{x^2+8x+15} \geq 0$ (15)

Solution: The key numbers are values of x where $x^2-1 = (x-1)(x+1) = 0$ from which $x = \pm 1$ and $x^2+8x+15 = (x+3)(x+5) = 0$ from which $x = -3$ and $x = -5$.
Thus the key numbers are $x = -5, -3, -1, 1$.
This yields the intervals $(-\infty, -5),\ (-5, -3),\ (-3, -1),\ (-1, 1)\ , (1, \infty)$
This yields

Interval	Test value	$x-1$	$x+1$	$x+5$	$x+3$	$\frac{(x+1)(x-1)}{(x+5)(x+3)}$
$(-\infty, -5)$	-6	-	-	-	-	+
$(-5, -3)$	-4	-	-	+	-	-
$(-3, -1)$	-2	-	-	+	+	+
$(-1, 1)$	0	-	+	+	+	-
$(1, \infty)$	2	+	+	+	+	+

Hence $\frac{x^2-1}{x^2+8x+15} > 0$ in the interval $(-\infty, -5) \cup (-3, -1) \cup (1, \infty)$. Since $\frac{x^2-1}{x^2+8x+15} = 0$ at $x = \pm 1$, then the solution set is $(-\infty, -5) \cup (-3, -1] \cup [1, \infty)$

4. The sum of the first n natural numbers is given by $1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$. For which values of n will the sum be less than 210? (10)

Solution: We want n such that $\frac{n(n+1)}{2} < 210$
$n(n+1) < 420$

$n^2 + n - 420 < 0$
$(n + 21)(n - 20) < 0$
So the critical numbers are $n = -21,\ n = 20$
So we get the intervals $(-\infty, -21),\ (-21, 20),\ (20, \infty)$
This yields

Interval	Test value	$(x + 21)$	$(x - 20)$	$(x + 21)(x - 20)$
$(\infty, -21)$	-22	-1	-1	+
$(-21, 20)$	0	+	-	-
$(20, \infty)$	21	+	+	+

There $\frac{n(n+1)}{2} < 210$ for all $n \in (-21, 20)$ but since natural numbers are positive, then we expect n to be positive hence this yields the solution set $(0, 20)$, i.e. for all natural